RideFlow Programming Languages and Technologies

Main programming languages

Language	Used for
TypeScript	Main frontend and backend application language
JavaScript	Runtime language underneath TypeScript and browser behavior
HTML	Web-page structure and the React entry document
CSS	Layout, colors, responsive design, themes, and animations
SQL	Database tables, migrations, queries, and data operations
Markdown	README files, technical guides, and documentation
JSON	Configuration files and structured data

Main frameworks and libraries

Technology	Role in RideFlow
React 19	Builds the interactive browser interface
Tailwind CSS 4	Utility-based styling and responsive layouts
Vite	Frontend development server and production build tool
Node.js	Runs TypeScript/JavaScript on the server
Express	HTTP web server and API host
tRPC 11	Typed communication between the React frontend and backend
Drizzle ORM	Maps TypeScript code to the SQL database
Vitest	Automated unit and server-procedure tests
Lucide React	Interface icons
Wouter	Frontend routing

Infrastructure and external services

Technology	Role
MySQL/TiDB	Stores users, profiles, fares, quotes, ledger entries, presence, and activity
S3-compatible storage	Stores profile photos and driver documents
OIDC/OAuth 2.0	Login and identity management
Stripe	Card payments and signed payment webhooks
Safaricom Daraja API	M-Pesa payments and asynchronous callbacks
Git	Tracks code history and changes
GitHub	Private source-code repository
Ubuntu/Linux	Recommended development and deployment operating system

Recommended learning order

1. Learn HTML and CSS fundamentals.
2. Learn JavaScript fundamentals: variables, functions, arrays, objects, promises, modules, and error handling.
3. Learn TypeScript: types, interfaces, unions, generics, and compiler errors.
4. Learn React: components, props, state, effects, forms, and conditional rendering.
5. Learn SQL and MySQL: tables, keys, indexes, joins, migrations, and transactions.
6. Learn Node.js and Express: servers, routes, middleware, environment variables, and HTTP responses.
7. Learn tRPC and Drizzle: typed procedures, authorization, queries, mutations, and schema mapping.
8. Learn Git and GitHub: branches, commits, pull requests, review, and rollback.
9. Learn authentication, private storage, payments, testing, monitoring, and deployment.

One-sentence project description

RideFlow is a TypeScript full-stack web application using React, Express, tRPC, Drizzle ORM, MySQL/TiDB, S3-compatible storage, OIDC authentication, Stripe, and Safaricom Daraja.